

An Overview of the Python Data Analysis Library

Pandas is a cross-platform library (abstraction) written in Python, Cython and C by Wes McKinney for the Python programming language. It is used for data analysis and data manipulation. This article lists a few important features of this library.



It is easy to install Pandas. Download Anaconda for your operating system and the latest Python version, and run the installer as available at https://pandas.pydata.org/getting_started.html. Pandas can be installed using pip from PyPI using the command `pip install pandas`.

You can verify the Python version by using the command `python --version`.

Output: Python 3.8.3

`pip list` and `pip freeze` will generate a list of installed packages:

```
1. pip freeze
2. anchorecli==0.8.2
3. astroid==2.4.2
4. .
5. .
6. .
7. openpyxl==3.0.5
8. .
9. pandas==1.1.3
10. .
11. py==1.8.2
12. .
13. .
14. .
15. xlrd==1.2.0
```

16. XlsxWriter==1.3.7

Table 1 lists some of the modules utilised in our script.

Module	Description
Pandas	Data import, merge, experimentation, clean-up, and analysis
OpenPyXL	Read/write Excel 2010 xlsx/xlsm files
xlrd	Read Excel data
XlsxWriter	Write to Excel (xlsx) files
matplotlib	Data visualisation
ConfigParser	Configuration file parser

Table 1: Module details

Figure 1 offers a quick look at Pandas features and Figure 2 gives a Pandas cheat sheet.



Figure 1: The features of Pandas